Proposition 5.1

parabolic rescaling Han--Li--Sun Proposition 5.1 (X_0, T)
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$$X_j(x,t) = \lambda_j\Big(X(x, t_j + \lambda_j^{-2}t) - X(x_j, t_j)\Big), \qquad \lambda_j = H(x_j, t_j).$$

Proposition 5.1 spacetime L^2
mean curvature flow shrinking sphere $|H_\infty|(0,0) = 1$ smooth pointed blow-up limit

Han--Li--Sun beta-symplectic gradient flow fixed-center Proposition 5.1

$$q_j = \lambda_j\big(F(x_j, t_j) - X_0\big), \qquad \theta_j = \lambda_j^2(T - t_j).$$

q_j θ_j H_j, V_j, f_j L^2 $F_j^\perp$ $q_j^\perp$ L^2
$q_j \to 0$ spacetime area bound Han--Li--Sun moving-cutoff
(5.5)

$F(x_j, t_j) \to X_0$ $q_j \to 0$

$$|F(x_j, t_j) - X_0| = o(\lambda_j^{-1});$$

essential Type-I $\lambda_j \asymp (T - t_j)^{-1/2}$ $o(\sqrt{T - t_j})$ $\lambda_j = |H|, |A|, (T - t_j)^{-1/2}$
q_j $q_j \to 0$ Chen--Li
4.7 q_j x_k p cutoff
$q = 0$

1.

$F : \Sigma \times [0, T) \to M$ beta-symplectic critical-surface gradient flow normal coordinates

$$\partial_\tau F = f, \qquad f = \frac{\cos^2\alpha\, H - \beta \sin^2\alpha\, V}{\cos^2\alpha + \beta\sin^2\alpha},$$

H mean-curvature vector V Han--Li--Sun normal field α Kähler angle symplectic-angle bound

$$\cos\alpha \geq \delta > 0.$$

Han--Li--Sun (X_0, T)

$$G_\lambda(x,u) = \lambda\big(F(x, T+\lambda^{-2}u) - X_0\big), \qquad u < 0.$$

Proposition 5.1 $R > 0$ $-\infty < a < b < 0$ $\lambda \to \infty$

$$\int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |H_\lambda|^2\, d\mu_u^\lambda\, du, \quad \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |V_\lambda|^2\, d\mu_u^\lambda\, du,$$
$$\int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |f_\lambda|^2\, d\mu_u^\lambda\, du, \quad \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |G_\lambda^\perp|^2\, d\mu_u^\lambda\, du.$$

369 $x_j \in \Sigma$ $t_j \nearrow T$

$$p_j = F(x_j, t_j), \qquad \lambda_j = |H|(x_j, t_j) \longrightarrow \infty,$$

$$F_j(x,t) = \lambda_j\Big(F(x, t_j + \lambda_j^{-2}t) - p_j\Big). \tag{1.1}$$

rescaled time interval

$$F_j(x_j, 0) = 0, \qquad |H_j|(x_j, 0) = 1. \tag{1.2}$$

(1.1) $R > 0$ $s_1 < s_2 < 0$ Proposition 5.1

2. scaling identities

$$\tau_j(t) = t_j + \lambda_j^{-2}t.$$

translation tangent normal spaces ambient dilation $y \mapsto \lambda_j y$ induced metric λ_j^2 Σ

$$g_j = \lambda_j^2 g, \qquad d\mu_t^j = \lambda_j^2 d\mu_{\tau_j(t)}, \qquad d\tau = \lambda_j^{-2} dt. \tag{2.1}$$

second fundamental form inverse metric

$$H_j = \lambda_j^{-1} H. \tag{2.2}$$

Kähler angle dilation scaled orthonormal tangent frame λ_j^{-1}

$$V_j = \lambda_j^{-1} V. \tag{2.3}$$

$$f_j = \frac{\cos^2\alpha_j\, H_j - \beta \sin^2\alpha_j\, V_j}{\cos^2\alpha_j + \beta\sin^2\alpha_j} = \lambda_j^{-1} f, \qquad \partial_t F_j = f_j. \tag{2.4}$$

$\cos\alpha_j \geq \delta$ uniform pointwise estimate

$$|f_j|^2 \leq C(\beta,\delta)\big(|H_j|^2 + |V_j|^2\big). \tag{2.5}$$

fixed singular point (X_0, T)

$$q_j = \lambda_j(p_j - X_0), \qquad \theta_j = \lambda_j^2(T - t_j). \tag{2.6}$$

fixed-center coordinates

$$G_j(x,t) := \lambda_j\big(F(x,\tau_j(t)) - X_0\big) = F_j(x,t) + q_j, \tag{2.7}$$

$$T - \tau_j(t) = \lambda_j^{-2}(\theta_j - t). \tag{2.8}$$

backward heat kernel

$$\rho_{X_0,T}(F,\tau_j(t))\, d\mu_{\tau_j(t)} = \psi_j(F_j,t)\, d\mu_t^j, \tag{2.9}$$

$$\psi_j(z,t) = \frac{1}{4\pi(\theta_j - t)} \exp\left(-\frac{|z+q_j|^2}{4(\theta_j - t)}\right). \tag{2.10}$$

normal-position identity

$$G_j^\perp = (F_j + q_j)^\perp = F_j^\perp + q_j^\perp. \tag{2.11}$$

fixed-center monotonicity $(F_j + q_j)^\perp$ $F_j^\perp$

3. Gaussian lower bound sharp criterion

$R > 0$ $s_1 < s_2 < 0$ cylinder

$$|F_j| \leq R, \qquad s_1 \leq t \leq s_2$$

j $c_R > 0$ $\psi_j \geq c_R$

$$\sup_j |q_j| < \infty, \qquad \sup_j \theta_j < \infty. \tag{3.1}$$

$|q_j| \leq Q$ $0 \leq \theta_j \leq \Theta$

$$-s_2 \leq \theta_j - t \leq \Theta - s_1, \qquad |F_j + q_j| \leq R + Q,$$

$$\psi_j(F_j,t) \ge \frac{1}{4\pi(\Theta - s_1)} \exp\left(-\frac{(R+Q)^2}{4(-s_2)}\right) > 0. \tag{3.2}$$

$\theta_j \to \infty$ $z=0, t=s_2$ prefactor $|q_j| \to \infty$ θ_j exponential factor ball (3.1) uniform Gaussian lower bound

4. “ ”

4.1 shrinking sphere

ordinary mean curvature flow beta-symplectic-specific counterexample “ 369 curvature-normalizing parametrization ”

$M_t = S^n_{r(t)} \subset \mathbb{R}^{n+1}$ shrinking round sphere

$$r(t) = \sqrt{2n(T-t)}.$$

e $t_j \nearrow T$ $p_j = r(t_j)e$

$$\lambda_j = |H|(p_j,t_j) = \frac{n}{r(t_j)}.$$

(1.1) j rescaled flow sphere center $-ne$ radius

$$r_j(t) = \sqrt{n^2 - 2nt},$$

$$|H_j|(t) = \frac{n}{\sqrt{n^2-2nt}}, \qquad |H_j|(x_j,0) = 1. \tag{4.1}$$

$\varepsilon > 0$ $s_1 = -\varepsilon$ $s_2 = -\varepsilon/2$

$$R > \sqrt{n^2 + 2n\varepsilon} - n,$$

$t \in [s_1,s_2]$ sphere $B_R(0)$ spherical cap

$$\int_{s_1}^{s_2}\int_{M_t^j \cap B_R(0)} |H_j|^2 \, d\mu_t^j \, dt$$

j Sun--Jun surface dimension $n=2$

4.2 smooth nonflat limit obstruction

round symmetry moving-center rescalings smooth locally converges F_∞ normalization

$$|H_\infty|(p_\infty, 0) = 1.$$

compact coordinate patch K $s_1 < s_2 < 0$

$$|H_\infty| \geq \frac{1}{2} \quad \text{on } K \times [s_1, s_2].$$

smooth convergence j $|H_j| \geq 1/4$ induced area density patch R $F_j(K,t) \subset B_R(0)$ $c > 0$

$$\liminf_{j\to\infty} \int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |H_j|^2 \, d\mu_t^j \, dt \geq c > 0. \tag{4.2}$$

beta-symplectic curvature-normalizing sequence smooth nonflat blow-up limit $|H_j|(x_j, 0) = 1$ compactness (4.2) regularity

5. beta-symplectic flow

Han--Li--Sun fixed-center Proposition 5.1

Theorem 5.1 moving-center transfer

F beta-symplectic gradient flow (X_0, T) fixed singular point $\lambda_j \to \infty$ (1.1) (2.6) F_j, q_j, θ_j

1. λ_j fixed-center rescalings

$$\widetilde{F}_j(x,u) = \lambda_j \big(F(x, T + \lambda_j^{-2} u) - X_0\big)$$

Han--Li--Sun Proposition 5.1 fixed-center spacetime L^2

2. $\sup_j |q_j| < \infty$ $0 \leq \theta_j \leq \Theta < \infty$
3. $[s_1, s_2] \subset (-\infty, 0)$ j moving rescaled time domain

$R > 0$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |H_j|^2 \, d\mu_t^j \, dt \to 0, \tag{5.1}$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |V_j|^2 \, d\mu_t^j \, dt \to 0, \tag{5.2}$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |f_j|^2 \, d\mu_t^j \, dt \to 0. \tag{5.3}$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |F_j^\perp|^2 \, d\mu_t^j \, dt \to 0, \tag{5.4}$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |q_j^\perp|^2 \, d\mu_t^j \, dt \to 0. \tag{5.5}$$

(5.5) normal-displacement condition Han--Li--Sun moving-cutoff (5.5)

$$q_j \to 0 \tag{5.6}$$

$A_R < \infty$

$$\int_{s_1}^{s_2} \mu_t^j\big(\Sigma_t^j \cap B_R(0)\big)\,dt \le A_R, \tag{5.7}$$

(5.4) moving-center

Proof

moving time t

$$u = t - \theta_j.$$

$\theta_j = \lambda_j^2(T - t_j)$

$$T + \lambda_j^{-2}u = t_j + \lambda_j^{-2}t.$$

$$\widetilde{F}_j(x,u) = F_j(x,t) + q_j. \tag{5.8}$$

translation H_j, V_j, f_j normal spaces induced measure $|F_j(x,t)| < R$ $|q_j| \le Q$

$$|\widetilde{F}_j(x,u)| < R + Q.$$

$0 \le \theta_j \le \Theta$ $t \in [s_1, s_2]$

$$u \in [s_1 - \Theta, s_2] \subset (-\infty, 0).$$

integrands

$$\int_{s_1}^{s_2}\int_{\{|F_j|<R\}} |H_j|^2 \le \int_{s_1-\Theta}^{s_2}\int_{\{|\widetilde{F}_j|<R+Q\}} |\widetilde{H}_j|^2.$$

fixed-center Proposition 5.1 (5.1) inclusion time shift (5.2) (5.3)

(2.5) (5.1) (5.2) (5.3)

normal-position term (2.11)

$$\widetilde{F}_j^{\perp} = F_j^{\perp} + q_j^{\perp}.$$

fixed-center Proposition 5.1 ball/time inclusion

$$\int_{s_1}^{s_2}\int_{\{|F_j|<R\}} |F_j^{\perp} + q_j^{\perp}|^2 \to 0. \tag{5.9}$$

$$|F_j^\perp|^2 \leq 2|F_j^\perp + q_j^\perp|^2 + 2|q_j^\perp|^2$$

(5.5) $\Rightarrow$ (5.4)

$$|q_j^\perp|^2 \leq 2|F_j^\perp + q_j^\perp|^2 + 2|F_j^\perp|^2$$

(5.9) (5.4) $\Rightarrow$ (5.5) orthogonal projection

$$\int |q_j^\perp|^2 \leq |q_j|^2 \int d\mu_t^j dt.$$

(5.6) (5.7)

6.

q_j θ_j shifted normal-position limit $F_j^\perp$ limit affine plane

$$F_j(u,v,t) = (u,v,-1) \subset \mathbb{R}^3, \qquad q_j = (0,0,1).$$

$H_j = V_j = f_j = 0$

$$F_j + q_j = (u,v,0)$$

tangent $(F_j + q_j)^\perp = 0$

$$F_j^\perp = (0,0,-1).$$

$R > 1$ $s_1 < s_2$

$$\int_{s_1}^{s_2} \int_{F_j \cap B_R(0)} |F_j^\perp|^2 \, d\mu \, dt = (s_2 - s_1)\pi(R^2 - 1) > 0.$$

center displacement normal component “ ”

7. $q_j \to 0$

X_0 normal coordinates

$$d_j = |F(x_j,t_j) - X_0|, \qquad q_j = \lambda_j(F(x_j,t_j) - X_0). \tag{7.1}$$

Proposition 7.1

$t_j \nearrow T$ $d_j \to 0$ $\lambda_j > 0$

$$q_j \to 0 \iff d_j = o(\lambda_j^{-1}). \tag{7.2}$$

$$|A|^2(x,t) \le \frac{K}{T-t}, \tag{7.3}$$

$\lambda_j = |A|(x_j,t_j)$ $\lambda_j = |H|(x_j,t_j)$

$$d_j = o(\sqrt{T-t_j}) \tag{7.4}$$

$q_j \to 0$ essential Type-I $c > 0$

$$\lambda_j \sqrt{T-t_j} \ge c, \tag{7.5}$$

(7.4) $\lambda_j = (T-t_j)^{-1/2}$ (7.4) $q_j \to 0$

Proof

$$|q_j| = \lambda_j d_j. \tag{7.6}$$

(7.2) Cauchy--Schwarz

$$|H|^2 \le 2|A|^2. \tag{7.7}$$

(7.3) $\lambda_j = |A|$ $\lambda_j\sqrt{T-t_j} \le \sqrt{K}$ $\lambda_j = |H|$ $\lambda_j\sqrt{T-t_j} \le \sqrt{2K}$ (7.6)

$$|q_j| = \big(\lambda_j\sqrt{T-t_j}\big)\frac{d_j}{\sqrt{T-t_j}},$$

(7.5) $|q_j| = d_j/\sqrt{T-t_j}$

7.2

o $d_j \to 0$ $T = 1$

$$F : S^2 \times [0,1) \longrightarrow \mathbb{R}^4, \qquad F(u,t) = 2\sqrt{1-t}\,u, \tag{7.8}$$

S^2 $\mathbb{R}^4$ $\partial_t F = H$

$$|A|^2 = \frac{1}{2(1-t)}, \qquad |H| = \frac{1}{\sqrt{1-t}}. \tag{7.9}$$

$t = 1$ $u_0 \in S^2$ $x_j = u_0$ $t_j \nearrow 1$ $X_0 = 0$ x_j

$$F(x_j,t_j) = 2\sqrt{1-t_j}\,u_0 \longrightarrow 0.$$

λ_j	$q_j = \lambda_j F(x_j,t_j)$
$\lvert H\rvert(x_j,t_j)$	$2u_0$
$\lvert A\rvert(x_j,t_j)$	$\sqrt{2}\,u_0$
$(1-t_j)^{-1/2}$	$2u_0$

(7.10)

mean curvature flow Han--Li--Sun beta-symplectic flow

“ $q_j \to 0$” beta-symplectic

(7.2) o

7.3

$q_j \to 0$ (7.4) $d_j = o(\lambda_j^{-1})$

$q_j \to 0$ Chen--Li q_j $q_j \to q$

8. Chen--Li 4.7

Chen--Li 305--308 Kähler--Einstein symplectic surface mean curvature flow

$$\lambda_k^2 = |A|^2(x_k, t_k) = \max_{t \le t_k} |A|^2, \qquad x_k \to p, \quad t_k \nearrow T, \tag{8.1}$$

$F(p,T) = X_0 = 0$

$$F_k(x,t) = \lambda_k\Big(F(x, t_k + \lambda_k^{-2}t) - F(p, t_k)\Big). \tag{8.2}$$

8.1 q_k $q_k \to 0$

$|H| \le \sqrt{2}|A|$

$$|F(p,t_k) - F(p,T)| \le \int_{t_k}^{T} |H(p,t)|\, dt \le C\sqrt{T - t_k}. \tag{8.3}$$

$\lambda_k\sqrt{T - t_k} \le C$

$$|q_k| := |\lambda_k F(p, t_k)| \le C. \tag{8.4}$$

$q_k \to q$ $q = 0$ 307 “we assume that $\lambda_k F(p, t_k) \to q$” q 4.7

8.2

(8.1)--(8.2)

$$|A_k|^2(x_k, 0) = 1, \qquad F_k(x_k, 0) = \lambda_k(F(x_k, t_k) - F(p, t_k)). \tag{8.5}$$

p

$$|A_k|^2(p, 0) = \lambda_k^{-2}|A|^2(p, t_k). \tag{8.6}$$

306 $|A_k|^2(p,0) \ge c > 0$ $x_k \to p$ $|A|^2(p, t_k) \ge c\lambda_k^2$ p x_k

$d(x_k, p)$ (8.6) p ambient ball

$$\lambda_k |F(x_k,t_k) - F(p,t_k)| \le C \tag{8.7}$$

(8.6) (8.7) smooth-limit $|A_\infty|(0,0) > 0$

8.3 cutoff

$s = t_k + \lambda_k^{-2} t$ $P_k = F(p,t_k)$ $d\mu_t^k = \lambda_k^2 d\mu_s$

$$F_k + \lambda_k P_k = \lambda_k F, \qquad t_k - s = -\lambda_k^{-2} t.$$

cutoff

$$\begin{aligned}
&\int_{\Sigma_t^k} \frac{1}{v}\, \phi_R(F_k) \frac{1}{-t} \exp\left(-\frac{|F_k + \lambda_k P_k|^2}{4(-t)}\right) d\mu_t^k \\
&= \int_{\Sigma_s} \frac{1}{v}\, \phi_R(\lambda_k(F - P_k))\, \frac{1}{t_k - s} \exp\left(-\frac{|F|^2}{4(t_k - s)}\right) d\mu_s.
\end{aligned} \tag{8.8}$$

306 $\phi_R(F)$ cutoff

$\chi \in C_c^\infty$ $\chi = 1$ X_0 $\lambda_k P_k$ rescaled ball $|Y| \le R$

$$F = P_k + \lambda_k^{-1} Y \longrightarrow X_0 \tag{8.9}$$

k $\chi(F) \equiv 1$ fixed original-coordinate cutoff $\phi_R(F_k)$ (8.8) k-dependent moving cutoff

8.4

monotonicity compactness smooth immersed surface $F_\infty : \Sigma_\infty \to \mathbb{R}^4$

$$H_\infty = 0, \qquad (F_\infty + q)^\perp = 0. \tag{8.10}$$

$\Omega = \{F_\infty + q \ne 0\}$ $F_\infty + q$ tangent vector $F_\infty + q = dF_\infty(W)$ tangent vector X

$$D_X(F_\infty + q) = dF_\infty(X)$$

normal component Gauss $A_\infty(X,W) = 0$ $e_1 = W/|W|$

$$A_\infty(e_1,e_1) = A_\infty(e_1,e_2) = 0.$$

$H_\infty = A_\infty(e_1,e_1) + A_\infty(e_2,e_2) = 0$ $A_\infty(e_2,e_2) = 0$ $A_\infty = 0$ Ω Ω $F_\infty + q$ F_∞ immersion $A_\infty \equiv 0$ q $q = 0$

8.5

compactness monotonicity $q = 0$

Theorem 8.1 q **blow-up contradiction .** $G_k : M_k \to \mathbb{R}^N$ rescaled time-slices $p_k \in M_k$

$$|A_k|(p_k) = 1, \qquad q_k \to q \in \mathbb{R}^N, \qquad \theta_k \to \theta \in (0,\infty). \tag{8.11}$$

U $p_\infty \in U$ compact exhaustion $K_i \Subset U$ K_i p_∞ embeddings $\Phi_{k,i} : K_i \to M_k$

$$\Phi_{k,i}(p_\infty) = p_k, \qquad G_k \circ \Phi_{k,i} \longrightarrow G_\infty \quad \text{in } C^2(K_i). \tag{8.12}$$

i $\varepsilon \in \{-1,1\}$ localized estimates

$$\int_{K_i} |H_k|^2 \, d\mu_k \longrightarrow 0, \tag{8.13}$$

$$\int_{K_i} \left| H_k + \frac{\varepsilon}{2\theta_k}(G_k + q_k)^\perp \right|^2 d\mu_k \longrightarrow 0, \tag{8.14}$$

contradiction q_k $q = 0$

Proof. (8.12) induced metrics area densities mean curvature vectors normal projections K_i (8.13)--(8.14)

$$H_\infty = 0, \qquad H_\infty + \frac{\varepsilon}{2\theta}(G_\infty + q)^\perp = 0 \quad \text{on } K_i^\circ. \tag{8.15}$$

K_i° U $\theta > 0$ $H_\infty = 0$ $(G_\infty + q)^\perp = 0$ U 8.4 $A_\infty \equiv 0$ (8.11)--(8.12) C^2 convergence

$$|A_\infty|(p_\infty) = \lim_{k\to\infty} |A_k|(p_k) = 1, \tag{8.16}$$

Theorem 8.1 Chen--Li rescaling $|A_k| = 1$ pointed compactness (8.8) fixed original-coordinate cutoff (8.13)--(8.14) p (8.7) x_k $G_k(x_k, 0) = 0$ Gaussian localized estimates x_k p

9. fixed cutoff

transfer proof fixed-center Proposition 5.1 ball inclusion time translation weighted monotonicity cutoff ϕ

$$\phi(F) = \phi\big(X_0 + \lambda_j^{-1}(F_j + q_j)\big)\,.$$

$|F_j| \le R$ $|q_j| \le Q$ $\lambda_j^{-1}(F_j + q_j) \to 0$ uniformly j cutoff (3.2)
Gaussian weight H_j, V_j unweighted estimates (2.5) f_j monotonicity square (2.11)
shifted normal-position estimate moving cutoff

10.

1. 369 curvature-normalizing moving-center scaling Han--Li--Sun fixed-center tangent-flow scaling
2. “ Proposition 5.1 ” shrinking sphere ordinary MCF
3. beta-symplectic flow beta-symplectic q_j, θ_j center-control hypotheses fixed-center proof moving-center version smooth nonflat normalized limit
4. q_j, θ_j fixed-center Proposition 5.1 $q_j^{\perp}$ L^2 $q_j \to 0$ local area bound
5. blow-up fixed-center scaling tangent behavior curvature-normalizing moving-center scaling nonflat singularity model
6. $F(x_j, t_j) \to X_0$ $q_j \to 0$ $\lambda_j \asymp (T - t_j)^{-1/2}$ sharp $|F(x_j, t_j) - X_0| = o(\sqrt{T - t_j})$
7. Chen--Li 4.7 q_k q $q = 0$ x_k p cutoff
8. pointed smooth compactness localized monotonicity $H_\infty = 0$ $(F_\infty + q)^{\perp} = 0$ $A_\infty = 0$ $q = 0$

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